1 Verilog Implementation

```
module phase2(
  input clk,
  input rst,
  input [3:0] code_in,
  output reg phase2_done,
  output reg phase2_fail
);
parameter safe=3'b000;
parameter s1=3'b001;
parameter done=3'b101;
parameter fail=3'b110;
reg alaram;
reg [2:0] current_state,next_state;
always @(posedge clk) begin
  if(rst)
    current_state=safe;
  else
    current_state=next_state;
end
always @(*) begin
  phase2_done=0;
  phase2_fail=0;
  alaram=0;
  case(current_state)
    safe:
      next_state=s1;
    s1:
      if(code_in == 4'b1101)
          next_state=done;
        else
          next_state=fail;
    done:
      phase2_done=1;
    fail: begin
      phase2_fail=1;
      alaram=1;
    end
    default:
      next_state=safe;
  endcase
end
endmodule
```